Limits of Cauchy approximations behave like limits for the closeness relation

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:RC-sim-lim' (source lines 1341-1343, Lemma 11.3.36); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 35) make the counted lemma print as Lemma 11.3.36, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: limits of Cauchy approximations behave like limits for closeness (Lemma 11.3.36).

Allowed standard prerequisites (editorial).

- Cauchy reals and their limits. RC is generated by rational points and by limits of Cauchy approximations, and closeness is the generated relation on them.. The proof inducts on RC and uses two of the closeness constructors explicitly.
- Roundedness and triangle inequality for closeness. closeness is rounded and satisfies the triangle inequality.. They are used at source lines 1351-1352.

Lemma 11.3.36. *For any $u : \mathbb{R}_{\mathsf{c}}$, Cauchy approximation y, and $\epsilon, \delta : \mathbb{Q}_+$, if $u \sim_\epsilon y_\delta$ then $u \sim_{\epsilon+\delta} \mathsf{lim}(y)$.*

Proof. We use $\mathbb{R}_{\mathsf{c}}$-induction on u. If u is $\mathsf{rat}(q)$, then this is exactly the second constructor of $\sim$. Now suppose u is $\mathsf{lim}(x)$, and that each x_η has the property that for any y, ϵ, δ, if $x_\eta \sim_\epsilon y_\delta$ then $x_\eta \sim_{\epsilon+\delta} \mathsf{lim}(y)$. In particular, taking $y :\equiv x$ and $\delta :\equiv \eta$ in this assumption, we conclude that $x_\eta \sim_{\eta+\theta} \mathsf{lim}(x)$ for any $\eta, \theta : \mathbb{Q}_+$.

Now let y, ϵ, δ be arbitrary and assume $\mathsf{lim}(x) \sim_\epsilon y_\delta$. By roundedness, there is a θ such that $\mathsf{lim}(x) \sim_{\epsilon-\theta} y_\delta$. Then by the above observation, for any η we have $x_\eta \sim_{\eta+\theta/2} \mathsf{lim}(x)$, and hence $x_\eta \sim_{\epsilon+\eta-\theta/2} y_\delta$ by the triangle inequality. Hence, the fourth constructor of $\sim$ yields $\mathsf{lim}(x) \sim_{\epsilon+2\eta+\delta-\theta/2} \mathsf{lim}(y)$. Thus, if we choose $\eta :\equiv \theta/4$, the result follows. □